

Study design and analytical workflow

Cohorts (role)

Development:

TCGA LGG/GBM, n = 667

External validation:

Gravendeel, n = 264 (main)

REMBRANDT, n = 342

(stress test)

Feature sets

A Clinical:

age, sex, grade, histology

B + Molecular:

IDH, 1p/19q, MGMT*

C Expression (exploratory):

40-gene panel; no imaging

Prediction target

Primary:

2-year overall survival

Sensitivity: 1- and 5-year

Robustness:

IPCW, 6-month landmark,

Cox / penalised Cox

Model families

Transparent:

base rate, age+grade LR,

penalised LR

Standard ML:

random forest, gradient boosting

Neural nets:

compact, medium, regularised MLP

Sensitivity: FT-Transformer

* where harmonisable

Validation

Internal:

repeated stratified CV in TCGA,

shared folds, out-of-fold probs

External:

train on full TCGA; test in

Gravendeel and REMBRANDT

(REMBRANDT: limited harmonisation)

Evaluation domains

Discrimination:

AUROC, AUPRC

Calibration:

Brier, intercept, slope

Clinical utility:

decision-curve net benefit

Comparison:

paired bootstrap (2000)

Prespecified criterion

Clinically meaningful, calibrated

improvement (all required):

- $\Delta\text{AUROC} \geq 0.03$
- no worse Brier score
- calibration slope 0.8–1.25